

ML Graded Quiz M7

Question 1

Load and prepare the cancer dataset as shown below:

```
from sklearn.datasets import load_breast_cancer

cancer = load_breast_cancer()

# Features

X = pd.DataFrame(cancer['data'], columns=cancer['feature_names'])

# Target label

y = pd.DataFrame(cancer['target'], columns=['Cancer'])

# Train test split

from sklearn.model_selection import train_test_split

import numpy as np

X_train, X_test, y_train, y_test = train_test_split(X, y,

! pip install scikit-optimize

from skopt import BayesSearchCV

# parameter ranges are specified by one of below

from skopt.space import Real
```

Define the support vector classifier SVC() hyperparameter space for "C" in the range Real(0.1, 1e+4) and "gamma" in the range Real(1e-6, 1e+1, 'log-uniform'). In our Bayes search space, define cv=5, random state = 42, n_iter=20 and verbose=3. Get the optimal hyperparameters for "C" and "gamma." From the resulting confusion matrix, what is the Recall of the model?

- ☐ 94%
- ☐ 89%
- ☐ 92%
- ☒ 90%

Question 2

Given the pipeline below:

```
from sklearn.preprocessing import StandardScaler

from sklearn.svm import SVC
```



QUESTION 1

- 1
- 2
- 5
- 6
- 9
- 10
- 13
- 14

```
from sklearn.pipeline import Pipeline

model = Pipeline([

    ('scaler', StandardScaler()),

    ('classifier', SVC())

])
```

How do we change the value of the hyperparameter "degree" to 2?

- ☐ model.set_params()['degree']==2
- ☐ model.set_parameters()['classifier__degree']==2
- ☐ model.set_params(classifier__degree=2)
- ☒ model.set_params(classifier__degree=5)

Question 3

Which of the following is a low variance model?

- ☐ K-nearest neighbor with $K = 1$
- ☐ Logistic regression
- ☐ Decision Trees
- ☒ Support vector machines

Question 4

Which of the following statements is **not** true?

- ☐ We can use Lasso regression in feature selection.
- ☐ Ridge regression reduces the model complexity and multicollinearity by shrinking the coefficients of the features.
- ☐ Both ridge and lasso regressions are used to prevent overfitting that may result from linear regression.
- ☒ Ridge and Lasso regression are used to prevent overfitting resulting from decision trees and random forest algorithms.

Question 5

Load and prepare the cancer dataset as shown below:

```
from sklearn.datasets import load_breast_cancer

cancer = load_breast_cancer()

# Features

X = pd.DataFrame(cancer['data'], columns=cancer['feature_names'])

# Target label
```

```

y = pd.DataFrame(cancer['target'], columns=['Cancer'])

# Train test split

from sklearn.model_selection import train_test_split

import numpy as np

X_train, X_test, y_train, y_test = train_test_split(X,

! pip install scikit-optimize

from skopt import BayesSearchCV

# parameter ranges are specified by one of below

from skopt.space import Real

```

How will we define a search space for an SVC model?

- ```

search_spaces = BayesianOptimization(bo_params_rf, {

'C': Real(0.1, 1

'gamma': Real

}))

```

- ```

search_spaces = {

'C': Real(0.1, 1e+4),

'gamma': Real(1e-6, 1e+1, 'log-uniform'),

}

```

- ```

search_spaces = {

'C', 'gamma'= [Real(0.1, 1e+4), Real(1e-6, 1e+1, 'log-unifo

}

```



```
search_spaces = BayesianOptimization(bo_params_rf, {

 'C'= Real(0.1, 1

 'gamma'= Real

 })
```

Question 6

Load the iris dataset below:

```
from sklearn.preprocessing import StandardScaler

from sklearn.ensemble import HistGradientBoostingClassifier

from sklearn.pipeline import Pipeline

from sklearn.model_selection import GridSearchCV

from sklearn.datasets import load_iris

X, y = load_iris(return_X_y=True)

pipeline = Pipeline([

 ('scaler', StandardScaler()),

 ('classifier', HistGradientBoostingClassifier(random_state=42))

])
```

How do we define the parameter grid for GridSearchCV?

- ☐ param\_grid = {'classifier\_\_learning\_rate': [0.1, 1, 10]}
- ☐ param\_grid = {'logistics\_\_learning\_rate': [0.1, 1, 10]}
- ☐ param\_grid = {'learning\_rate': [0.1, 1, 10]}
- ☒ param\_grid = {'HistGradientBoostingClassifier\_\_learning\_rate': [0.1, 1, 10]}

Question 7

If we choose a hyperparameter value for a model using 5-fold cross validation, what is the best strategy of choosing the final model and estimating the model error?

- ☐ Train the model on the entire dataset with the parameter we picked and use the average CV error to approximate the model error.
- ☐ Find the average of the 5 models we built and use the average CV error to approximate error.
- ☐ Find the average of the 5 models we built and use the average error the combined model gives to approximate the error.
- ☒ Randomly choose one of the 5 models we used in cross validation and approximate the error using the validation set.

Question 8

Which of the following statements is **true** about the linear regression regularization parameters?

- ☐ A decrease in the regularization parameter causes the model to underfit the data, and hence, the validation error becomes bigger.
- ☐ An increase in the regularization parameter causes the model to overfit the data, and hence, the validation error becomes bigger.
- ☐ An increase in the regularization parameter causes the model to underfit the data, and hence, the validation error becomes bigger.
- ☒ An decrease in the regularization parameter causes the model to underfit the data, and hence, the training error becomes bigger.

Question 9

Given the confusion matrix below, what is the accuracy of the model?

|                 | Predicted Negative | Predicted Positive |
|-----------------|--------------------|--------------------|
| Actual Negative | 115                | 6                  |
| Actual Positive | 14                 | 125                |

- ☐ 90%
- ☐ 89%
- ☐ 92%
- ☒ 95%

Question 10

Load the iris dataset below:

```
from sklearn.datasets import load_iris

iris = load_iris()

X = iris.data

y = iris.target
```

Inspect the target variable and choose the correct statement below.

- ☐ The class counts are imbalanced.
- ☐ The target variable is skewed to the right.
- ☐ The class counts are balanced.
- ☒ The target variable is skewed to the left.

Question 11

What is the danger of tuning hyperparameters on the test dataset?

- ☐ The model will underfit on the test dataset.

- ☐ The model will overfit on the training dataset.
- ☐ The model will give errors when used on the training dataset.
- ☒ The model will overfit on the test dataset.

Question 12

Which of these is the default score parameter in scikit learn when performing regression?

- ☐ Mean Absolute Error
- ☐ Mean Squared Error
- ☐ Median Absolute Error
- ☒  $R^2$

Question 13

How does Bayesian optimization work?

- ☐ Bayes optimization uses the past evaluation results to sample new candidates that are most likely to give better results.
- ☐ Bayesian optimization defines the probability distributions in which the hyperparameters belong and randomly picks the paired combination of hyperparameters.
- ☐ Bayesian optimization tries out all the possible combinations of parameters in a hyperparameter space.
- ☒ Bayesian optimization tests only as many tuples as you choose in a hyperparameter space.

Question 14

Load the cancer dataset below:

```
import pandas as pd

from sklearn.datasets import load_breast_cancer

from sklearn.svm import SVC

from sklearn.model_selection import train_test_split

import numpy as np

cancer = load_breast_cancer()

df_X = pd.DataFrame(cancer['data'], columns=cancer['feature_names'])

df_y = pd.DataFrame(cancer['target'], columns=['Cancer'])

Train test split
```

```
X_train, X_test, y_train, y_test = train_test_split(df,
])
```

By using GridSearchCV having verbose=3 and refit=True, what will be the best estimator of the hyperparameters "C" and "gamma" if their hyperparameter space are [0.1, 1, 10, 100, 1000] and [1, 0.1, 0.01, 0.001, 0.0001], respectively?

- ☐ {'C': 100, 'gamma': 0.0001}
- ☐ {'C': 100, 'gamma': 0.01}
- ☐ {'C': 0.1, 'gamma': 0.0001}
- ☒ {'C': 0.1, 'gamma': 1}

Question 15

Which of these statements describes the k-fold cross validation strategy?

- ☐ The model trains on the whole dataset  $k$  times and averages the accuracy.
- ☐ The model trains on  $k - 2$  folds, validates on 1 of the remaining set, and tests on the last set.
- ☐ The model trains on  $k/2$  folds if  $k$  is even or  $(k + 1)/2$  folds for odd  $k$  and validates on the rest of the left out set.
- ☒ The model prediction function is trained on  $k - 1$  folds and tested on the left out set.

Question 16

Which of these is a property of the baseline model?

- ☐ It should be based on the actual data.
- ☐ It should be based on the test data.
- ☐ It should be medium in complexity.
- ☒ It should be experimental, and therefore, an explanation of what the model does is not our priority.

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